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SENSING FROST AND ICE ACCUMULATION USING CAPACITANCE

Abstract

A capacitance sensing system senses frost and ice accumulation in an energy efficient defrost system. The capacitance sensing system comprises a first capacitor including a portion of a metal heat exchanger and a sensor electrode electrically isolated from the metal heat exchanger; a tank oscillator including a second capacitor and an inductor coupled in parallel with each other and with the first capacitor; and a circuit coupled to the tank oscillator. The circuit determines a resonant frequency of the tank oscillator, determines a capacitance value of the first capacitor based on the resonant frequency of the tank oscillator, and transmits a heater activation command in response to determining the capacitance value is greater than a threshold.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional of U.S. Pat. No. 17,750,615, filed May 23, 2022, which is a divisional of U.S. patent application Ser. No. 15/936,059 filed Mar. 26, 2018 (now U.S. Pat. No. 11,371,954), issued Jun. 28, 2022, which claims priority to U.S. Provisional Patent Application No. 62/552,911 filed Aug. 31, 2017, titled “AUTOMATED CONTINUOUS MONITORING OF FROST AND ICE FORMATION ON HEAT EXCHANGERS,” all of which are hereby incorporated herein by reference in their entireties.

BACKGROUND

[0002] Refrigeration equipment provides cooling by heat exchange between cold outer surfaces of a heat exchanger with a fluid that is passed over the outer surface of the heat exchanger—for example, relatively warmer air. The warmer fluid loses energy to the relatively colder heat exchanger, thereby cooling the fluid. When the fluid is air, the eventuality of water vapor in the air condensing on the outer surface of the heat exchanger and forming frost and/or ice is possible.

SUMMARY

[0003] In an example, a capacitance sensing system comprises a first capacitor including a portion of a metal heat exchanger and a sensor electrode electrically isolated from the metal heat exchanger; a tank oscillator including a second capacitor and an inductor coupled in parallel with each other and coupled in parallel with the first capacitor; and a circuit coupled to the tank oscillator. The circuit is configured to: determine a resonant frequency of the tank oscillator; determine a capacitance value of the first capacitor based on the resonant frequency of the tank oscillator; determine when the capacitance value is greater than a threshold; and transmit a heater activation command in response to determining the capacitance value is greater than the threshold.

[0004] In an example, a refrigerator with an energy efficient defrost system is provided. The refrigerator comprises a metal heat exchanger including an electrode of a first capacitor; a sensor electrode of the first electrode spaced apart from the metal heat exchanger; and a heater configured to be activated and deactivated based on a capacitance value of the first capacitor compared to a threshold capacitance, the capacitance value of the first capacitor being indicative of frost or ice accumulation in the refrigerator.

[0005] In an example, a circuit comprises a capacitor coupled between first and second terminals of the circuit; an inductor coupled between the first and second terminals in parallel with the capacitor; an adjustable frequency oscillator coupled to the first and second terminals; and a capacitance-to-digital converter; wherein the circuit is configured to determine a capacitance value of an external sensing capacitor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] For a detailed description of various examples, reference will now be made to the accompanying drawings in which:

[0007] FIG. 1 shows a system for sensing frost and ice accumulation by capacitance sensing in accordance with various examples;

[0008] FIG. 2 shows a heat exchanger and capacitance sensor electrode in accordance with various examples;

[0009] FIG. 3 shows a heat exchanger and another capacitance sensor electrode in accordance with various examples;

[0010] FIG. 4 shows a capacitance sensing circuit in accordance with various examples;

[0011] FIG. 5 shows another system for sensing frost and ice accumulation by capacitance sensing in accordance with various examples;

[0012] FIG. 6 shows yet another system for sensing frost and ice accumulation by capacitance sensing in accordance with various examples;

[0013] FIG. 7 shows a graph of capacitance versus time in accordance with various examples;

[0014] FIG. 8 shows a flow chart of a method in accordance with various examples;

[0015] FIG. 9 shows a system for sensing frost and ice accumulation including a capacitance sensing circuit having multiple channels of capacitance sensing in accordance with various examples; and

[0016] FIG. 10 shows another system for sensing frost and ice accumulation including a capacitance sensing circuit having multiple channels of capacitance sensing in accordance with various examples.

DETAILED DESCRIPTION

[0017] Heat exchangers in refrigeration equipment accumulate frost and ice that interferes with efficient heat exchange between the surfaces of the heat exchanger and air flow. This frost and ice results from condensation of water vapor present in the air flowing over the heat exchanger's surfaces followed by freezing of the water droplets. As frost and ice builds up, cooling of the air by the heat exchanger's surfaces becomes progressively less efficient. In some systems a heater is provided which melts off the frost and ice accumulation. Some systems perform heat cycles in an open loop mode based on time, which results in either executing heating cycles prematurely or unnecessarily or executing heating cycles too infrequently. In either case, the result is diminished refrigeration or cooling efficiency. Other systems sense temperature changes over time and trigger a heating cycle when a change in temperature exceeds a predefined threshold. The temperature changes, however, are so slight that achieving the desired precision measurements to rely on this feedback as a heat cycle triggering mechanism is a challenge.

[0018] The present disclosure teaches establishing a capacitor having one plate provided by a portion of the surface of a metal heat exchanger and a second plate provided by an electrically isolated electrode positioned proximate to the metal heat exchanger. This capacitor can be characterized by a capacitance value that varies over time based on a changing dielectric strength of the capacitor, where this dielectric strength is a function of the amount of water, frost, and/or ice on the portion of the surface of the metal heat exchanger that provides one of the plates of this capacitor. By determining the capacitance value of this capacitor with high accuracy, a level of ice, frost, and/or water accumulation on the metal heat exchanger can be inferred. This capacitance value can further be used to initiate a heating cycle to melt and remove the accumulated frost and ice and to turn off the heating cycle. For example, as the sensed capacitance exceeds a capacitance threshold, the system turns on the heater. As the frost and ice melts, it turns into water. Water has a significantly higher dielectric constant than frost and ice. The sensed capacitance first rapidly increases a significant amount as frost and ice becomes water, and the sensed capacitance rapidly decreases a significant amount as water drains off the metal heat exchanger surfaces. Various criteria or rules can be defined for deciding when to turn off the heater.

[0019] FIG. 1 shows an illustrative capacitance sensing system **100** for sensing frost and ice accumulation by capacitance sensing in accordance with various examples. The system comprises a refrigeration unit **102**. In examples, the unit **102** comprises a metal heat exchanger **104** and a heater **106**. In examples the metal heat exchanger **104** has metal surfaces but comprises other components that are not metal. In examples, the refrigeration unit **102** is a refrigerator, a freezer, or an air conditioner. When turned on, the heater **106** melts frost and ice that have accumulated on the surfaces of the metal heat exchanger **104**. A capacitor **108** is formed by a portion of the surface of

the metal heat exchanger **104** and a capacitance sensor electrode **110**. The capacitance sensor electrode **110** takes a variety of forms. In examples, the capacitance sensor electrode **110** is located in close proximity to the surface of the metal heat exchanger **104**. In examples, the capacitance sensor electrode **110** is planar in form. In examples, the capacitance sensor electrode **110** is sleeve-like in form. The system **100** further comprises a subsystem **112** that is configured to determine the capacitance value of the capacitor **108**, compare the capacitance value to a predefined threshold, and turn the heater **106** on when the capacitance value meets or exceeds the predefined threshold. The subsystem **112** determines the capacitance value in a variety of ways. In examples, the subsystem **112** determines the capacitance value of the capacitor **108** by determining a resonant frequency of a tank oscillator **410** that includes the capacitor **108**. In examples, the subsystem **112** determines the capacitance value of the capacitor **108** by charging the capacitor **108** to a first known voltage and discharge the capacitor **108** through a known resistance to a ground, measure the time it takes to discharge from the first known voltage to a second known voltage and determine the capacitance value based on these known values. In examples, the subsystem **112** determines the capacitance value of capacitor **108** using other methods of measuring capacitance. [0020] In examples, the subsystem **112** further turns off the heater **106** when a criteria or rule is satisfied. For example, the subsystem **112** turns off the heater **106** after a predefined time interval has run since turning on the heater **106**. For example, the subsystem **112** turns off the heater **106** after the capacitance value rises to a maximum value and begins to decrease. For example, the subsystem **112** turns off the heater **106** after the capacitance value rises to a maximum value, decreases rapidly to a lower capacitance value, and then maintains a stable capacitance value. In examples, other criteria are used for triggering turning off the heater **106**.

[0021] FIG. 2 shows an illustrative capacitor **200**. The capacitor **200** comprises a portion of a surface of a metal heat exchanger **202** that is electrically connected to ground **204**. In examples, the ground **204** is a connection to a chassis of an enclosing system, such as a refrigeration unit. In examples, the ground **204** is a connection to a common wire in the enclosing system, such as a refrigeration unit. The capacitor **200** further comprises a capacitance sensor electrode **206** that is formed from a mesh of wires (e.g., a mesh of interconnected wires). In examples the mesh of wires comprises metal. In examples the mesh of wires comprises copper, stainless steel, and/or aluminum. The capacitance sensor electrode **206** is connected to an external lead **208** that promotes connecting to external circuitry (not explicitly shown).

[0022] FIG. 3 shows an illustrative capacitor **300**. The capacitor **300** comprises a portion of a surface of a metal heat exchanger **302** that is electrically connected to ground **304**. In examples, the ground **304** is a connection to a chassis of an enclosing system, such as a refrigeration unit. In examples, the ground **304** is a connection to a common wire in the enclosing system, such as a refrigeration unit. The capacitor **300** further comprises a capacitance sensor electrode **306** that is formed by perforating a sheet of metal and then stretching the perforated sheet of metal. In examples, the perforated sheet of metal comprises copper, stainless steel, and/or aluminum. The capacitance sensor electrode **306** is connected to an external lead **308** that promotes connecting to external circuitry (not explicitly shown). It is desirable that the capacitance sensor electrode **206**, **306** are open and provide free flow through of air.

[0023] FIG. 4 shows an illustrative capacitance sensing circuit **400**. The circuit **400** comprises a first capacitor **402** (e.g., such as the capacitors described in FIGS. 1-3), and a tank oscillator **410**. The tank oscillator **410** comprises a second capacitor **404**, an inductor **406**, and a variable frequency oscillator **408**. The first capacitor **402**, the second capacitor **404**, and the inductor **406** are all in parallel with each other. The resonant frequency of the capacitance sensing circuit **400** depends on the inductance value of the inductor **406** and on the aggregate capacitance of the first capacitor **402** in parallel with the second capacitor **404**. The first capacitor **402** is shown separated from the right-hand of the circuit to suggest that the first capacitor **402** can be located remotely from the remainder of the circuit **400**. This remote location does not change the nature of the

function of this circuit **400**. The frequency of the variable frequency oscillator **408** can be varied or swept across a range of frequencies to find the resonant frequency of the capacitance sensing circuit. In examples an integrated circuit separate from the capacitance sensing circuit **400** controls the variable frequency oscillator **408** to sweep a frequency input to the tank oscillator across a range of frequencies.

[0024] Because the capacitance of the second capacitor **404** is known, the inductance of the inductor **406** is known, and the frequency of the oscillator **408** is known as it is swept across a range of frequencies, the unknown capacitance value of the first capacitance can be determined based on the known values of resonant frequency, inductance, and capacitance of the second capacitor **404**. More specifically, the capacitance of the first capacitor can be determined from the equation for resonant frequency:

$$[00001] F = \frac{1}{2\sqrt{LC}} \quad \text{EQ1}$$

where F is the resonant frequency, L is the inductance value of the inductor **406**, and C is the aggregate capacitance of the first capacitor **402** and the second capacitor **404**. This equation can be transformed to find the capacitance of the first capacitor as:

$$[00002] C_1 = \frac{(\frac{1}{2F})^2}{L} - C_2 \quad \text{EQ2}$$

where F is the resonant frequency, L is the inductance value of the inductor **406**, C.sub.1 is the capacitance value of the first capacitor **402**, C.sub.2 is the capacitance value of the second capacitor **404**, and

$$C = C_{\text{sub.1}} + C_{\text{sub.2}}.$$

[0025] FIG. 5 shows another illustrative system **500** for sensing frost and ice accumulation by capacitance sensing in accordance with various examples. The system **500** comprises a refrigeration unit **502** that comprises a metal heat exchanger **504** and a heater **506**. A first capacitor **508** is formed by a portion of a surface of the metal heat exchanger **504** and a capacitance sensor electrode, although the capacitor **508** is represented here with a capacitor symbol for ease of understanding.

[0026] The first capacitor **508** is coupled to a capacitance sensing circuit **510**. A tank oscillator **410** comprises a second capacitor **520**, inductor **522**, and an adjustable frequency oscillator **524**. The second capacitor **520**, the inductor **522**, and the adjustable frequency oscillator **524** are all connected in parallel with each other and in parallel with the first capacitor **508**. The cut-line between the first capacitor **508** and the capacitance sensing circuit **510** is used to illustrate that the coupling can occur over a distance. In examples, the distance separating the first capacitor **508** from the capacitance sensing circuit **510** ranges from one inch to one foot to ten feet. The capacitance sensing circuit **510** comprises a capacitance-to-digital converter **526** and the tank oscillator **410**. In examples, the capacitance-to-digital converter **526** is an integrated circuit. In examples, the adjustable frequency oscillator **524** is integrated within the capacitance-to-digital converter **526**.

[0027] The capacitance-to-digital converter **526** varies the frequency of the adjustable frequency oscillator **524** to find the resonant frequency of the tank oscillator **410** and the first capacitor **508**. Based on the resonant frequency and based on the known inductance of the inductor **522**, the capacitance-to-digital converter **526** determines the capacitance of the parallel combination of the first capacitor **508** and the second capacitor **520**. In examples, the capacitance-to-digital converter **526** converts the capacitance of the parallel combination of the first capacitor **508** and the second capacitor **520** to a digital representation and outputs this digital representation of this aggregate capacitance value to a processor **512**. In other examples, the capacitance-to-digital converter **526** determines the capacitance value of the first capacitor **508**, digitizes that value, and outputs this digital representation of the capacitance of the first capacitor **508** to the processor **512**.

[0028] In examples, the capacitance sensing circuit **510** is implemented as a printed circuit board

(PCB). In examples, the capacitance sensing circuit **510** comprises a plurality of independent capacitance sensing channels, each comprising a capacitor in series with an inductor coupled to a frequency variable oscillator. Said in other words, in examples, the capacitance sensing circuit **510** comprises a plurality of the tank oscillators **410**, and each different tank oscillator is coupled to a different channel of the capacitance-to-digital converter **526** (see FIG. **9** and associated text description hereinafter). In examples, multiple independent capacitance sensing channels share a single frequency variable oscillator. In examples, each of multiple independent sensing channels is associated with its own independent frequency variable oscillator. In examples, the inductor **522** is implemented as a planar spiral inductor. In examples, the inductor **522** is implemented as a planar spiral multilayer inductor. In examples, the capacitance sensing circuit **510** is able to provide a capacitance resolution of 200 femtofarads or less, 100 femtofarads or less, 50 femtofarads or less, or 10 femtofarads or less.

[0029] In examples, the processor **512** is a microcontroller (MCU) integrated circuit, a microprocessor (MPU) integrated circuit, a field programmable gate array (FPGA) integrated circuit, a complex programmable logic device (CPLD) integrated circuit, or an application specific integrated circuit (ASIC) integrated circuit. The processor **512** compares the digital representation of capacitance received from the capacitance-to-digital converter **526** and/or from the capacitance sensing circuit **510** to a predefined threshold value of capacitance. When the received capacitance value exceeds the predefined threshold value of capacitance, the processor **512** sends a heater turn on command to the heater **506**. The processor **512** subsequently sends a heater turn off command to the heater **506** when a heater turn-off criterion is satisfied. The heater turn-off criterion may be referred to as a predefined heater turn-off criterion in some contexts. The heater turn-off criterion may be built into the logic that the processor **512** executes.

[0030] In examples, the processor **512** continues to monitor the digital representation of capacitance received from the capacitance sensing circuit **510** and sends a heater turn-off command to the heater **506** based on its monitoring of the digital representation of capacitance received. In examples, the processor **512** sends the heater turn-off command in response to monitoring the capacitance value increasing to a maximum and then beginning to decrease. In examples, the processor **512** sends the heater turn-off command in response to monitoring the capacitance value increasing to a maximum, decreasing to a minimum, and then becoming stable close to that minimum. In some contexts, a heater turn-on command is referred to as a heater activation command and a heater turn-off command is referred to as a heater deactivation command. In examples, the predefined heater turn-off criterion is the capacitance value increasing to a maximum and then beginning to decrease. In examples the predefined heater turn-off criterion is monitoring the capacitance value increasing to a maximum, decreasing to a minimum, and then becoming stable close to that minimum.

[0031] In examples, the determining of capacitance enables the system to determine when to turn the heater on and off so that cycles of heating the metal heat exchanger **504** are minimized, conserving energy and also reducing unnecessary audible noise that is associated with the heat cycles. Additionally, in examples, eliminating unnecessary heating cycles and/or reducing duration of heating cycles reduces stress on the mechanisms of the refrigeration unit **502**, thereby extending the unit's life.

[0032] FIG. **6** shows yet another illustrative system **600** for sensing frost and ice accumulation by capacitance sensing in accordance with various examples. The system **600** comprises a refrigeration unit **602** that comprises a metal heat exchanger **604** and a heater **606**. A first capacitor **608** is formed by a portion of a surface of the metal heat exchanger **604** and a capacitance sensor electrode.

[0033] The first capacitor **608** is coupled to a printed circuit board (PCB) **610**. The PCB **610** comprises a tank oscillator **410**. The tank oscillator **410** comprises a second capacitor **620**, an inductor **622**, and an adjustable frequency oscillator **624**. The second capacitor **620**, the inductor

622, and the adjustable frequency oscillator **624** are all connected in parallel with each other and in parallel with the first capacitor **608**. The cut-line between the first capacitor **608** and the PCB **610** is used to suggest that the coupling can occur over a distance. In examples, the distance separating the first capacitor **608** from the PCB **610** ranges from one inch to one foot to ten feet. The PCB **610** further comprises a capacitance-to-digital converter **626**. In examples, the capacitance-to-digital converter **626** is an integrated circuit. In examples, the adjustable frequency oscillator **624** is integrated within the capacitance-to-digital converter **626**.

[0034] The capacitance-to-digital converter **626** varies the frequency of the adjustable frequency oscillator **624** to find the resonant frequency of the tank oscillator **410** and the first capacitor **608**. Based on the resonant frequency and based on the known inductance of the inductor **622**, the capacitance-to-digital converter **626** determines the capacitance of the parallel combination of the first capacitor **608** and the second capacitor **620**. In examples, the capacitance-to-digital converter **626** converts the capacitance of the parallel combination of the first capacitor **608** and the second capacitor **620** to a digital representation and outputs this digital representation of this aggregate capacitance value to a processor **628**. In other examples, the capacitance-to-digital converter **626** determines the capacitance value of the first capacitor **608**, digitizes that value, and outputs this digital representation of the capacitance of the first capacitor **608** to the processor **628**.

[0035] In examples, the PCB **610** comprises a plurality of independent capacitance sensing channels including a capacitor in series with an inductor coupled to a frequency variable oscillator. In examples, multiple independent capacitance sensing channels share a single frequency variable oscillator. In examples, each of multiple independent sensing channels is associated with its own independent frequency variable oscillator (see FIG. **10** and associated text description hereinafter). In examples, the inductor **622** is implemented as a planar spiral inductor. In examples, the inductor **622** is implemented as a planar spiral multilayer inductor. In examples, the PCB **610** is able to provide a capacitance resolution of 200 femtofarads or less, 100 femtofarads or less, 50 femtofarads or less, or 10 femtofarads or less.

[0036] In examples, the processor **628** is a microcontroller integrated circuit, a microprocessor integrated circuit, a field programmable gate array (FPGA) integrated circuit, a complex programmable logic device (CPLD) integrated circuit, or an application specific integrated circuit (ASIC) integrated circuit. The processor **628** compares the digital representation of capacitance received from the capacitance-to-digital converter **626** and/or from the PCB **610** to a predefined threshold value of capacitance. When the received capacitance value exceeds the predefined threshold value of capacitance, the processor **628** sends a heater turn-on command to the heater **606**. The processor **628** subsequently sends a heater turn-off command to the heater **606** when a criterion is satisfied. In examples, the processor **628** continues to monitor the digital representation of capacitance received from the PCB **610** and sends a heater turn-off command to the heater **606** based on its monitoring of the digital representation of capacitance received. In examples, the processor **628** sends the heater turn-off command in response to monitoring the capacitance value increase rapidly to a relatively high maximum and then begin to decrease. In examples, the processor **628** sends the heater turn-off command in response to monitoring the capacitance value increase rapidly to a relatively high maximum, decrease sharply to a minimum, and then become stable close to that minimum. The processor **628** executes instructions or logic to determine the control signal sent to the heater **606** based on the digital representation of capacitance received from the capacitance sensing circuit **612**. In examples, the processor **628** is located off of the PCB **610**. In other examples, the processor **628** is located on the PCB **610**.

[0037] FIG. **7** shows an illustrative graph **700** of capacitance versus time in accordance with various examples. The graph **700** has an X-axis **702** or independent variable axis that corresponds to time and a Y-axis **704** or dependent variable axis that corresponds to capacitance. Over time frost and ice accumulate on the metal heat exchanger **104** of FIG. **1**, which causes a sensed capacitance associated with the capacitor **108** formed by the portion of the surface of the metal heat exchanger

104 and the capacitance sensor electrode **110** to increase due to the increased dielectric constant associated with accumulated frost and ice. The sensed capacitance over time is shown as trace **706**. At a predefined threshold value of capacitance **710**, the heater **106** is commanded to be turned on, the frost and ice melt, the capacitance goes up rapidly to a peak value at **712** due to conversion of frost and ice to water which has a much higher dielectric constant than frost and ice. As heating continues, the capacitance then decreases rapidly toward the capacitance **714** as the water drains off of the surface of the metal heat exchanger **104** until the capacitance establishes a steady level **716** corresponding to a water-free, frost-free, ice-free surface of the metal heat exchanger **104**.

[0038] In some examples, the steady level **716** of the capacitance of capacitor **108** varies slowly over time, for example over months or years. This variance can occur due to oxidation on a surface of the sensor electrode **110** and/or oxidation on the surface of the metal heat exchanger **104** Notwithstanding, in examples, the threshold **708** is automatically adapted, based on the slowly changing value of the steady level **716**, to modulate heat cycles for defrosting and deicing the metal heat exchanger **104**. Said in other words, the minimum capacitance value that occurs after turning off the heater is determined and deemed a clean or non-ice perturbed capacitance value. This clean capacitance value is then used to adapt the threshold **708**. In examples, a specific increase in capacitance is associated with a corresponding thickness of ice and frost accumulation on the metal heat exchanger. Thus, in examples, the threshold **708** is adapted to have the value of this specific increase in capacitance added to the minimum capacitance value.

[0039] FIG. **8** shows an illustrative flow chart of a method **800** in accordance with various examples. In examples, the method **800** is performed by the processor **512**, **628**. At block **802** a capacitance value is received. In examples, the capacitance value represents the capacitance value of the first capacitor **508**, **608**. In examples, the capacitance value represents the sum of the capacitance values of the first capacitor **508**, **608** added with the capacitance value of the second capacitor **520**, **620**. The change in the sum of the capacitance values of the first capacitor **508**, **608** added with the capacitance value of the second capacitor **520**, **620** will change based largely on the change of the capacitance value of the first capacitor **508**, **608** alone and hence this aggregate capacitance value still provides a reference against a threshold for triggering activation of the heater **506**, **606**.

[0040] At block **804**, the received capacitance value is compared to a predefined threshold. If the received capacitance value is less than the predefined threshold, the processing returns to block **802**. If the received capacitance value does exceed the predefined threshold, processing proceeds to block **806**. At block **806**, the heater is commanded to turn on.

[0041] At block **810**, a capacitance value is received. The capacitance value that is received while the heater remains turned on is evaluated to see if a criteria or rule has been satisfied. A variety of criteria or rules are contemplated by the present disclosure. In examples, the criteria or rules specify that after the received value of capacitance manifests a rapid increase to a maximum and then drops, turn off the heater at that point or turn off the heater after a short time delay. In examples the criteria or rules specify that after the received value of capacitance manifests a rapid increase to a maximum, then drops to a value less than the predefined threshold for turning on the heater, and then achieves a stable value, then turn off the heater. After turning the heater off at block **814**, the processing returns to block **802**. The processing of method **800** continues from a power-on of the processor **512**, **628** to power-off.

[0042] FIG. **9** shows another illustrative system **900** for sensing frost and ice accumulation by capacitance sensing in accordance with various examples. The system **900** is substantially similar to the system **500** described above with reference to FIG. **5**, except that in FIG. **9** a capacitance sensing circuit **910** comprises multiple separate tank oscillators, each coupled to a separate external capacitance. While four separate tank oscillators are shown in FIG. **9** as part of the capacitance sensing circuit **910**, in examples a different number of separate tank oscillators are part of the capacitance sensing circuit **910**.

[0043] The system **900** comprises the capacitance sensing circuit **910**. The capacitance sensing circuit **910** comprises a first tank oscillator **912**, a second tank oscillator **914**, a third tank oscillator **916**, and a fourth tank oscillator **918**. The first tank oscillator **912** comprises a capacitor **922**, an inductor **924**, and an oscillator **926**; the second tank oscillator **914** comprises a capacitor **932**, an inductor **934**, and an oscillator **936**; the third tank oscillator **916** comprises a capacitor **942**, an inductor **944**, and an oscillator **946**; the fourth tank oscillator **918** comprises a capacitor **952**, an inductor **954**, and an oscillator **956**. In examples, the capacitance sensing circuit **910** is a printed circuit board.

[0044] An external capacitance **902** associated with a heat exchanger is coupled in parallel to the first tank oscillator **912**, and a heater **972** is configured to defrost the heat exchanger associated with the external capacitance **902**. An external capacitance **904** associated with a heat exchanger is coupled in parallel to the second tank oscillator **914**, and a heater **974** is configured to defrost the heat exchanger associated with the external capacitance **904**. An external capacitance **906** associated with a heat exchanger is coupled in parallel to the third tank oscillator **916**, and a heater **976** is configured to defrost the heat exchanger associated with the external capacitance **906**. An external capacitance **908** associated with a heat exchanger is coupled in parallel to the fourth tank oscillator **918**, and a heater **978** is configured to defrost the heat exchanger associated with the external capacitance **908**. Each of the external capacitances **902**, **904**, **906**, **908** is formed by a plate and a metal surface of a heat exchanger body, as described above with reference to FIG. 1, FIG. 2, and FIG. 3.

[0045] The capacitance sensing circuit **910** comprises a capacitance-to-digital converter (CDC) **920**. In examples, the CDC **920** is an integrated circuit. The CDC **920** varies the frequency of the oscillators **926**, **936**, **946**, **956**, each in turn, to find the resonant frequency of its associated tank oscillator **912**, **914**, **916**, **918** and capacitor **902**, **904**, **906**, **908**. In examples, the oscillators are not located external to the CDC **920** but instead are located within the CDC **920**. In examples, the CDC **920** comprises a single oscillator and the CDC **920** connects the single oscillator to each of the tank oscillators **912**, **914**, **916**, **918** one at a time in sequence, for example via a multiplexer. Based on the resonant frequencies of each tank oscillator **912**, **914**, **916**, **918** and based on the known inductances of the inductors **924**, **934**, **944**, **954**, the CDC **920** determines the capacitances of the parallel combinations of the capacitors **902** and **922**, of the capacitors **904** and **932**, of the capacitors **906** and **942**, and of the capacitors **908** and **954**.

[0046] The CDC **920** outputs these capacitance values as a digital representation to a processor **960**. In examples, the processor **960** is a microcontroller (MCU) integrated circuit, a microprocessor (MPU) integrated circuit, a field programmable gate array (FPGA) integrated circuit, a complex programmable logic device (CPLD) integrated circuit, or an application specific integrated circuit (ASIC) integrated circuit. The processor **960** compares the digital representations of the capacitance values received from the capacitance-to-digital converter **920** and/or from the capacitance sensing circuit **910** to a predefined threshold value of capacitance. When one of the received capacitance values exceeds a predefined threshold value of capacitance, the processor **960** sends a heater turn on command to the associated heater. In examples, different threshold values of capacitance may be predefined by the processor **960** for different ones of the capacitances **902**, **904**, **906**, **908**. The processor **960** is coupled by a first heater turn on line **962** to the heater **972**, by a second heater turn on line **964** to the heater **974**, by a third heater turn on line **966** to the heater **976**, and by a fourth heater turn on line **968** to the fourth heater **978**.

[0047] FIG. 10 shows another illustrative system **1000** for sensing frost and ice accumulation by capacitance sensing in accordance with various examples. The system **1000** is substantially similar to the system **900** described above with reference to FIG. 9, except that in FIG. 10 the capacitance sensing circuit **1010** comprises a processor **1060**. In an example, the capacitance sensing circuit **1010** is a printed circuit board. The system **1000** comprises multiple separate tank oscillators each coupled to a separate external capacitance. While four separate tank oscillators are shown in FIG.

10 as part of the capacitance sensing circuit **1010**, in examples a different number of separate tank oscillators are part of the capacitance sensing circuit **1010**.

[0048] The system **1000** comprises a capacitance sensing circuit **1010**. The capacitance sensing circuit comprises a first tank oscillator **1012**, a second tank oscillator **1014**, a third tank oscillator **1016**, and a fourth tank oscillator **1018**. The first tank oscillator **1012** comprises a capacitor **1022**, an inductor **1024**, and an oscillator **1026**; the second tank oscillator **1014** comprises a capacitor **1032**, an inductor **1034**, and an oscillator **1036**; the third tank oscillator **1016** comprises a capacitor **1042**, an inductor **1044**, and an oscillator **1046**; the fourth tank oscillator **1018** comprises a capacitor **1052**, an inductor **1054**, and an oscillator **1056**.

[0049] An external capacitance **1002** associated with a heat exchanger is coupled in parallel to the first tank oscillator **1012**, and a heater **1072** is configured to defrost the heat exchanger associated with the external capacitance **1002**. An external capacitance **1004** associated with a heat exchanger is coupled in parallel to the second tank oscillator **1014**, and a heater **1074** is configured to defrost the heat exchanger associated with the external capacitance **1004**. An external capacitance **1006** associated with a heat exchanger is coupled in parallel to the third tank oscillator **1016**, and a heater **1076** is configured to defrost the heat exchanger associated with the external capacitance **1006**. An external capacitance **1008** associated with a heat exchanger is coupled in parallel to the fourth tank oscillator **1018**, and a heater **1078** is configured to defrost the heat exchanger associated with the external capacitance **1008**. Each of the external capacitances **1002**, **1004**, **1006**, **1008** is formed by a plate and a metal surface of a heat exchanger body, as described above with reference to FIG. 1, FIG. 2, and FIG. 3.

[0050] The capacitance sensing circuit **1010** comprises a capacitance-to-digital converter (CDC) **1020**. In examples, the CDC **1020** is an integrated circuit. The CDC **1020** varies the frequency of the oscillators **1026**, **1036**, **1046**, **1056**, each in turn, to find the resonant frequency of its associated tank oscillator **1012**, **1014**, **1016**, **1018** and capacitor **1002**, **1004**, **1006**, **1008**. In examples, the oscillators are not located external to the CDC **1020** but instead are located within the CDC **1020**. In examples, the CDC **1020** comprises a single oscillator and the CDC **1020** connects the single oscillator to each of the tank oscillators **1012**, **1014**, **1016**, **1018** one at a time in sequence, for example via a multiplexer. Based on the resonant frequencies of each tank oscillator **1012**, **1014**, **1016**, **1018** and based on the known inductances of the inductors **1024**, **1034**, **1044**, **1054**, the CDC **1020** determines the capacitances of the parallel combinations of the capacitors **1002** and **1022**, of the capacitors **1004** and **1032**, of the capacitors **1006** and **1042**, and of the capacitors **1008** and **1054**.

[0051] The CDC **1020** outputs these capacitance values as a digital representation to a processor **1060**. In examples, the processor **1060** is a microcontroller (MCU) integrated circuit, a microprocessor (MPU) integrated circuit, a field programmable gate array (FPGA) integrated circuit, a complex programmable logic device (CPLD) integrated circuit, or an application specific integrated circuit (ASIC) integrated circuit. The processor **1060** compares the digital representations of the capacitance values received from the capacitance-to-digital converter **1020** and/or from the capacitance sensing circuit **1010** to a predefined threshold value of capacitance. When one of the received capacitance values exceeds a predefined threshold value of capacitance, the processor **1060** sends a heater turn on command to the associated heater. In examples, different threshold values of capacitance may be predefined by the processor **1060** for different ones of the capacitances **1002**, **1004**, **1006**, **1008**. The processor **1060** is coupled by a first heater turn on line **1062** to the heater **1072**, by a second heater turn on line **1064** to the heater **1074**, by a third heater turn on line **1066** to the heater **1076**, and by a fourth heater turn on line **1068** to the fourth heater **1078**.

[0052] In the foregoing discussion and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . .” Also, the term “couple” or “couples” is intended to mean either an indirect or direct wired

or wireless connection. Thus, if a first device couples to a second device, that connection can be through a direct connection or through an indirect connection via other devices and connections. [0053] The above discussion is meant to be illustrative of the principles and various embodiments of the present disclosure. Numerous variations and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.

Claims

1. A refrigerator comprising: a metal heat exchanger including an electrode of a first capacitor; a sensor electrode of the first electrode spaced apart from the metal heat exchanger; and a heater configured to be activated and deactivated based on a capacitance value of the first capacitor compared to a threshold capacitance, the capacitance value of the first capacitor being indicative of frost or ice accumulation in the refrigerator.
 2. The refrigerator of claim 1, further comprising: a tank oscillator including a second capacitor, an inductor and an adjustable frequency oscillator, the tank oscillator having a first and second terminals between which each of the second capacitor, the inductor and the first capacitor are coupled; and a capacitance-to-digital converter coupled to the tank oscillator.
 3. The refrigerator of claim 2, wherein the inductor is planar spiral inductor.
 4. The refrigerator of claim 2, further comprising: a processor coupled to the capacitance-to-digital converter and to the heater, wherein the capacitance-to-digital converter is configured to determine at least one of the capacitance value of the first capacitor and an aggregate capacitance value of the first and second capacitors and to output an indication of at least one of the capacitance value and the aggregate capacitance value to the processor.
 5. A circuit comprising: a capacitor coupled between first and second terminals of the circuit; an inductor coupled between the first and second terminals in parallel with the capacitor; an adjustable frequency oscillator coupled to the first and second terminals; and a capacitance-to-digital converter; wherein the circuit is configured to determine a capacitance value of an external sensing capacitor.
 6. The circuit of claim 5, wherein the capacitance-to-digital converter controls the adjustable frequency oscillator to generate output signals, each having a different frequency, to identify a resonant frequency of the circuit and to determine the capacitance value of the external sensing capacitor based on the resonant frequency.
 7. The circuit of claim 5, wherein the inductor is planar spiral inductor.
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